ROV FOR AQUATIC INTEGRITY

An AI-Ready Underwater Robot for Water Monitoring and Emergency Support

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Abstract— This paper presents a low-cost remotely operated vehicle (ROV) that combines underwater mobility, visual inspection, water-quality sensing, and operator-centered AI alerts. The prototype adapts an open 3D-printable ROV concept into a four-layer architecture built around ESP32 and ESP32-CAM controllers, a tethered RS485 data link, three brushless A2212 thrusters, and pH, total dissolved solids (TDS), and temperature sensing. Rather than treating artificial intelligence as an autonomous authority, the system uses anomaly and person-detection modules to prioritize observations while retaining video verification and human control. The assembled prototype demonstrates the feasibility of a modular educational platform; however, waterproofing, sensor calibration, model training, and controlled field trials remain necessary before quantitative performance claims can be made.
Keywords—remotely operated vehicle; water quality; ESP32; sensor fusion; underwater robotics; human-in-the-loop AI.

1. INTRODUCTION

Water monitoring and aquatic emergency response often require personnel to approach remote, contaminated, or physically hazardous locations. Handheld meters provide local measurements but cannot reposition themselves; underwater cameras improve visibility but may not integrate environmental sensors; and basic low-cost ROVs provide mobility without an analytic alert layer. The proposed ROV for Aquatic Integrity addresses this fragmentation by placing propulsion, imaging, sensing, data transmission, and decision support on one modular platform.

The design goal is not full autonomy. It is a practical, inspectable system that extends an operator's reach while keeping consequential interpretation under human supervision. The vehicle streams water parameters and imagery to a surface station, reports motor and link status, and generates risk cues that the operator confirms against live video. This approach is appropriate for an early prototype because it separates sensing and alerting from final action, reducing the danger of over-trusting incomplete training data.

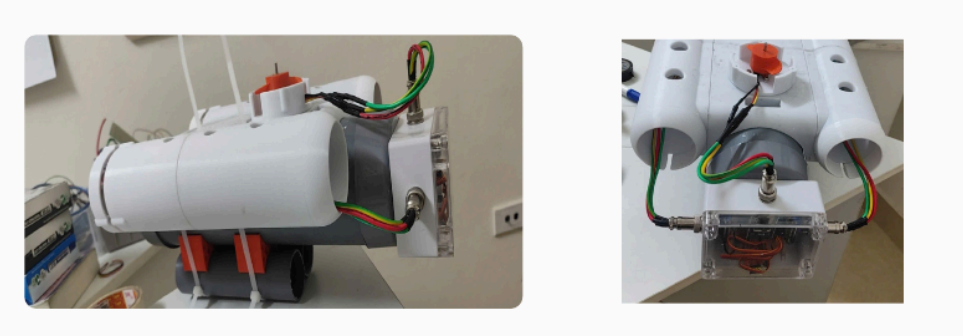


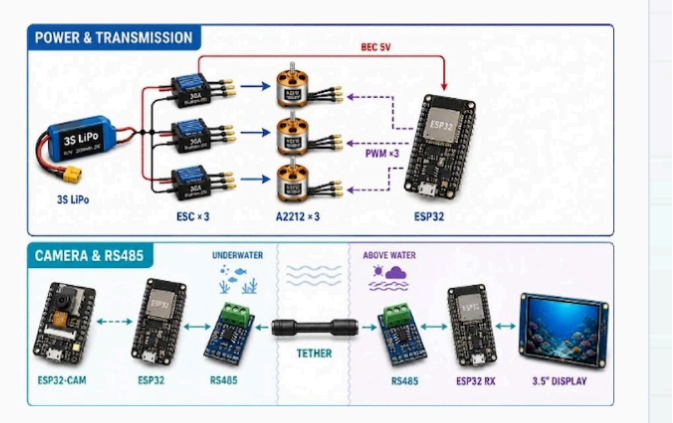
Fig. 1. Side and front views of the assembled 3D-printed ROV prototype, reproduced from the project poster [1].

2. DESIGN BASIS AND SYSTEM ARCHITECTURE

The mechanical starting point is "The Albie," an open 3D-printed ROV published by mfilion in 2018 [1]. That reference emphasizes a deliberately simple vehicle for underwater exploration, using three submerged brushless motors, reversible electronic speed controllers (ESCs), a tethered video path, and a watertight electronics enclosure. The present project retains the accessible, printable form factor while replacing the original radio/video concept with an ESP32-centered sensing and telemetry stack. This is an adaptation rather than a direct duplication: the structural precedent lowers fabrication cost, whereas the project's contribution lies in integrated water sensing, surface data display, modular processing, and AI-ready alert logic.

2.1 Four-Layer Model

The architecture is organized into four layers. Layer 1 contains power and physical signals: a 3S lithium-polymer battery, a 5 V battery-eliminator circuit (BEC), three PWM channels, RS485, the camera, and water sensors. Layer 2 performs embedded analysis and connection monitoring. Layer 3 fuses observations, assigns alert states, and transmits operator commands through the tether. Layer 4 presents the dashboard, video, water data, motor status, and warnings. The partition is useful because individual subsystems can be tested without requiring the complete vehicle to be operational.



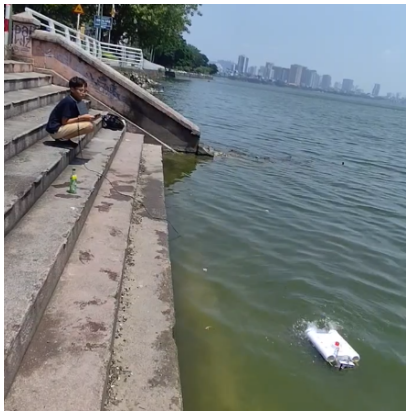


Fig. 2. Project architecture and ESP32-based electrical/data pathways [1].

2.2 Processing Modules

Module	Primary role	Inputs	Output / location
WaterGuard	Water-quality anomaly detection	pH, TDS, temperature	Risk score / ESP32
PersonFusion	Sensor fusion for possible person detection	Radar, PIR, camera	Person score / ESP32
VisionConfirm	Image-based confirmation	ESP32-CAM image	Confirmation / ESP32-CAM
MotorControl	Three-axis propulsion control	3 × PWM	3 ESCs and A2212 motors
DataLink	Tether telemetry and display	RS485 data	Surface ESP32 and 3.5-in LCD

Table 1. Functional decomposition derived from the project poster [1].

3. PROTOTYPE IMPLEMENTATION

The prototype uses an ESP32 as the principal controller and an ESP32-CAM for image capture. A 3S LiPo battery supplies the propulsion chain, while a 5 V BEC regulates power for the microcontrollers. Three ESCs receive independent PWM commands and drive three A2212 brushless motors, enabling horizontal maneuvering and vertical depth control. The tether carries RS485 telemetry to a surface ESP32 receiver, where a 3.5-inch display can present operating data. In the intended configuration, pH, TDS, and temperature are sampled into a common record so that changes can be time-aligned with imagery and vehicle status [1].

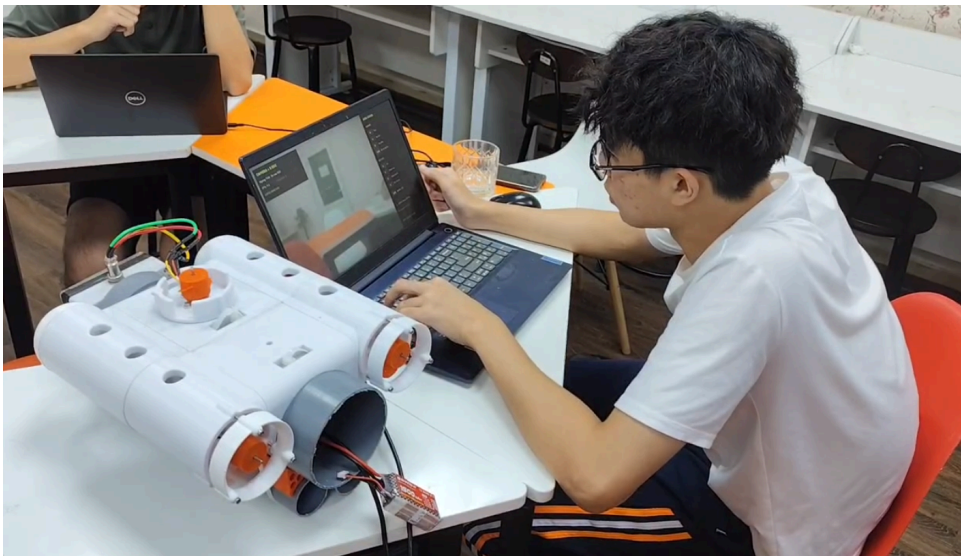


Fig. 3. Team member working beside the printed ROV frame and open electronics enclosure; project media folder [3].

3.1 Human-in-the-Loop Intelligence

The proposed intelligence is divided into WaterGuard, PersonFusion, and VisionConfirm. WaterGuard converts the water-sensor stream into an anomaly score and risk level. PersonFusion combines radar, passive-infrared, and camera evidence into a person score, while VisionConfirm supplies image-based corroboration. The paper treats these as an

integration design and development target, not as fully validated models. No training-set size, accuracy, latency, false-alarm rate, or confusion matrix is provided in the current project materials. Accordingly, the responsible claim is that the hardware and software interfaces are AI-ready, while model performance remains to be established through labeled data collection and testing.

3.2 Feasibility and Preliminary Results

The available evidence supports three preliminary results. First, the printable vehicle body has been assembled and physically integrates propulsion mounts, an electronics container, and wiring routes. Second, the control concept connects the ESP32, three ESC channels, brushless motors, camera, and tethered surface display in a modular electrical architecture. Third, the selected components are widely available and replaceable, which makes the platform suitable for iterative school-level engineering. Photographs in the Drive folder document fabrication and assembly activity, while the poster records completed subsystem integration [1], [3].

These results demonstrate build feasibility but not yet operational validity in natural water. Waterproof integrity, buoyancy and trim, tether drag, endurance, sensor drift, communication reliability, and propulsion performance require controlled measurement. The absence of quantitative test data is therefore a limitation of the current stage rather than evidence of failure. It defines the next experimental program.

4. VALIDATION PLAN AND DISCUSSION

Validation should proceed from safety-critical component tests to integrated field trials. A dry electrical inspection should verify polarity, current limits, ESC arming behavior, and emergency shutdown. The sealed enclosure should then undergo a static immersion test without powered electronics, followed by a pressure-equivalent or depth-stepped leak test. Water sensors should be calibrated against reference solutions and a trusted thermometer; repeated measurements should quantify bias, repeatability, warm-up time, and drift. Tether communication should be stressed with packet-loss and latency logging while motors operate, because switching noise may degrade sensor and video data.

After bench validation, the vehicle should be tested in a pool or tank using defined trajectories, depth holds, and obstacle approaches. Metrics should include mission duration, power consumption, command-to-motion latency, video availability, RS485 error rate, temperature rise inside the enclosure, and recovery behavior after link interruption. AI modules should be evaluated separately on labeled datasets before live alerts are enabled. WaterGuard requires normal and abnormal water profiles; PersonFusion and VisionConfirm require diverse positive and negative scenes. Thresholds should be selected from precision–recall trade-offs, and the interface should expose confidence and sensor availability rather than only a binary alarm.

4.1 Limitations, Safety, and Ethics

The prototype must not be represented as a certified rescue device or as a substitute for laboratory-grade water analysis. Low-cost pH and TDS probes are sensitive to calibration, fouling, temperature compensation, and electrical interference. Camera-based detection can fail in turbid water, poor lighting, occlusion, or unfamiliar scenes. Batteries and high-current propulsion components introduce fire, short-circuit, and entanglement risks. Testing should use a retrieval line, propeller guards, a surface observer, protected battery handling, and an immediate power-isolation method. Any captured imagery should be stored only as long as necessary for engineering evaluation and handled with appropriate consent and privacy safeguards.

5. CONCLUSION

ROV for Aquatic Integrity transforms a simple 3D-printable underwater vehicle concept into an integrated platform for water-quality monitoring, visual inspection, tethered control, and human-supervised alerting. Its strongest contribution at the present stage is architectural: sensing, propulsion, imaging, communication, and decision support are separated into modules that can be built and verified incrementally. The assembled prototype and documented electronics show practical feasibility, while the explicit operator-verification principle provides a sound safety boundary. The next milestone is not greater autonomy, but disciplined validation—waterproofing, calibration, reliability testing, labeled data collection, and reproducible field trials.

REFERENCES

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